FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS | GRADE 10 | SUB-STRAND 1.3: QUADRATIC EQUATIONS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This Final Explanation document is your opportunity to show everything you have learned across the 7 lessons of this unit. Your task is to answer the driving question fully and clearly:

"How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai?"

Read each section prompt carefully before you write. Use correct mathematical vocabulary and show all working. Your answer should tell a complete story — starting from why a quadratic equation was needed, moving through how you solved it, and ending with what your solution means for the Githurai community. A strong Final Explanation does not just give answers; it explains the reasoning behind every step, connects mathematics to the real world, and shows that you truly understand what quadratic equations are, where they come from, and why they matter.

Write in full sentences where asked. Label all diagrams. Show all algebraic working step by step. Check every answer by substituting back into the original equation or context.

Section 1: Setting Up the Problem — From Real Life to a Quadratic Equation

The Githurai Harambee committee needs a rectangular hall with an area of exactly 120 m². The length of the hall must be 2 metres more than the width. **(a) Let the width of the hall be w metres. Write an expression for the length in terms of w.**	(a) If the width is w metres, then the length is $(w + 2)$ metres. (b) Area = length × width $120 = (w + 2) \times w$ $120 = w^2 + 2w$ Rearranging: $w^2 + 2w - 120 = 0$ ✓ (c) This problem cannot be solved with a linear equation because the unknown w is multiplied by itself ($w \times w = w^2$), producing a squared term. A linear equation only contains variables raised to the power of 1. The highest power of w in this equation is 2, which

(b) Write an equation using the formula for the area of a rectangle. Show how this equation simplifies to the standard form of a quadratic equation: $w^2 + 2w - 120 = 0$. **(c) Explain in 2–3 sentences why this problem cannot be solved using a simple linear equation. What feature of the equation makes it quadratic rather than linear?**	is the defining feature of a quadratic equation — it means the relationship between w and the area is curved, not straight, and we need quadratic methods to find the exact value of w.

Section 2: Solving by Factorisation	
Solve the equation $w^2 + 2w - 120 = 0$ by factorisation. **(a) Find two numbers that multiply to −120 and add to +2. Show your reasoning.** **(b) Write the fully factorised form of the equation and solve for w.** **(c) You will get two solutions. Explain clearly why one solution must be rejected, and state the valid value of w with its units.** **(d) What is the length of the hall? Show how you found it.**	(a) We need two numbers whose product is −120 and whose sum is +2. Testing factor pairs of 120: $10 \times 12 = 120$, and $12 - 10 = 2$ ✓ So the two numbers are +12 and −10. (b) $w^2 + 2w - 120 = 0$ $(w + 12)(w - 10) = 0$ Either $w + 12 = 0 \rightarrow w = -12$ Or $w - 10 = 0 \rightarrow w = 10$ (c) A width cannot be a negative number — you cannot have a hall that is −12 metres wide. Negative solutions are mathematically valid but have no meaning in this real-world context. Therefore $w = -12$ is rejected. The valid solution is $w = 10$ metres. (d) Length = $w + 2 = 10 + 2 = 12$ metres. The hall is 10 m wide and 12 m long. Check: $10 \times 12 = 120$ m² ✓

Section 3: Solving by the Quadratic Formula (Alternative Method)	
The quadratic formula can solve any quadratic equation of the form $ax^2 + bx + c = 0$: **$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$**	(a) Comparing $w^2 + 2w - 120 = 0$ with $aw^2 + bw + c = 0$: $a = 1$, $b = 2$, $c = -120$ (b) $w = (-2 \pm \sqrt{2^2 - 4 \times 1 \times (-120)}) / (2 \times 1)$ $w = (-2 \pm \sqrt{4 + 480}) / 2$

(a) Identify the values of a, b, and c in the equation $w^2 + 2w - 120 = 0$. **(b) Substitute these values into the quadratic formula and solve for w. Show every line of working.** **(c) Do your answers match the solutions from Section 2? What does this tell you about the two methods?** **(d) Explain what the discriminant ($b^2 - 4ac$) tells us about the number of solutions. What would it mean for the hall project if the discriminant were negative?**	$w = (-2 \pm \sqrt{484}) / 2$ $w = (-2 \pm 22) / 2$ Solution 1: $w = (-2 + 22) / 2 = 20 / 2 = 10$ Solution 2: $w = (-2 - 22) / 2 = -24 / 2 = -12$ (c) Yes — both methods give $w = 10$ and $w = -12$. This shows that factorisation and the quadratic formula are equivalent methods. When factorisation is straightforward, it is faster; the formula is useful when factors are not obvious or when solutions are not whole numbers. (d) The discriminant is $b^2 - 4ac = 484 > 0$, which means the equation has two distinct real solutions. If the discriminant were negative, there would be no real solutions — meaning no real width exists that gives that exact area. For the hall project, this would mean the committee's area target is mathematically impossible with the given length–width relationship, and they would need to revise their plans.

Section 4: Connecting Quadratic Equations to Graphs	
The quadratic equation $w^2 + 2w - 120 = 0$ is related to the quadratic function $f(w) = w^2 + 2w - 120$. **(a) Explain what the solutions (roots) of the equation represent on the graph of this function.** **(b) Sketch the graph of $f(w) = w^2 + 2w - 120$. Label: the two roots, the y-intercept, and the vertex (turning point). You may calculate the vertex using $w = -b/2a$.** **(c) The committee later considers a second design where the area must be at least 120 m². Shade the region on your graph that represents widths for which $f(w) < 0$. What does this region tell the committee about acceptable**	(a) The solutions of the equation $w^2 + 2w - 120 = 0$ are the values of w where the graph of f(w) crosses the horizontal axis (the w-axis). These are called the roots or zeros of the function. On the graph, the two roots appear at $w = -12$ and $w = 10$. (b) Key features: – Roots (x-intercepts): $w = -12$ and $w = 10$ – y-intercept: $f(0) = 0 + 0 - 120 = -120$ → point $(0, -120)$ – Vertex: $w = -b/2a = -2/2 = -1$ $f(-1) = (-1)^2 + 2(-1) - 120 = 1 - 2 - 120 = -121$ Vertex: $(-1, -121)$ – The parabola opens upward ($a = 1 > 0$). [Sketch: an upward-opening parabola crossing the w-axis at −12 and 10, with vertex at (−1, −121) and y-intercept at −120.] (c) The region where $f(w) < 0$ is between the two roots: $-12 < w < 10$. Since width must be positive, the relevant portion is $0 < w < 10$. This tells the committee that any width less than 10 m (with length = w + 2) gives an area less than 120 m². To meet or exceed the 120 m² requirement, the width must be at least 10 m — confirming that w = 10 m is the minimum acceptable width for their design.

widths?	

Section 5: Answering the Driving Question — Mathematics in the Community	
Now bring everything together to answer the driving question in full. **(a) Summarise how quadratic equations differ from linear equations, using the Githurai hall as your example.** **(b) Describe the complete solution to the hall problem: state the dimensions you found, verify them, and explain what they mean for the community's building project.** **(c) Give ONE other example of a real planning or community problem in Kenya where a quadratic equation would be more useful than a linear equation. Describe the situation, define a variable, and write (but do not solve) the quadratic equation it would produce.** **(d) Reflect: Before this unit, you tried to find the hall's width by trial and error or guessing. How has learning quadratic equations changed the tools available to you? Why does this matter beyond the classroom?**	(a) A linear equation contains the unknown variable raised only to the power of 1 (e.g., $2w + 3 = 23$), and its graph is a straight line. A quadratic equation contains the variable raised to the power of 2 as the highest power (e.g., $w^2 + 2w - 120 = 0$), and its graph is a parabola. In the Githurai hall problem, the area formula Area = $w(w + 2)$ naturally produces a squared term because we are multiplying two lengths together — making a quadratic equation unavoidable. No linear method can find the exact answer. (b) Using factorisation: $w^2 + 2w - 120 = 0 \rightarrow (w + 12)(w - 10) = 0 \rightarrow w = 10$ (rejecting $w = -12$). Width = 10 m, Length = 12 m. Verification: $10 \times 12 = 120$ m² ✓ For the community in Githurai, this means the construction team can now draw accurate plans, order the correct amount of materials, and ensure the hall meets its design requirement. A mathematical solution replaces guesswork with certainty. (c) Example — Designing a maize storage shed in Kericho: A farmers' cooperative in Kericho wants to fence a rectangular storage area using exactly 40 metres of fencing, and they need the area to be 96 m². Let the width be x metres. Then the length is $(20 - x)$ metres [since $2x + 2l = 40$, so $l = 20 - x$]. Area: $x(20 - x) = 96$ $20x - x^2 = 96$ $x^2 - 20x + 96 = 0$ [This quadratic equation would then be solved to find the exact dimensions of the shed.] (d) Before this unit, finding the hall's width meant guessing values and testing them one by one — a slow process with no guarantee of finding the exact answer. Learning to form and solve quadratic equations gives us a reliable, efficient, and precise method that works every time. Beyond the classroom, this matters because engineers designing bridges in Nairobi, architects planning estates in Kisumu, and farmers calculating optimal plot sizes in the Rift Valley all face problems where quantities multiply together — and quadratic equations are the mathematical tool that unlocks those answers with certainty.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Problem Formulation — Setting up the quadratic equation from the real-world context	Correctly defines the variable, writes both expressions (width and length), applies the area formula accurately, and	Correctly sets up the equation and reaches standard form with minor arithmetic slips. Explains the quadratic	Attempts to write an equation but makes errors in forming expressions or applying the area formula. May reach an incorrect

	rearranges to standard form $w^2 + 2w - 120 = 0$ with full working shown. Clearly explains why the problem is quadratic and not linear, using precise mathematical vocabulary (squared term, highest power, degree).	nature of the problem adequately, though explanation may lack full mathematical precision.	standard form or struggle to explain the difference between linear and quadratic equations.
Algebraic Solution — Solving by factorisation and/or the quadratic formula	Solves accurately by both factorisation and the quadratic formula, showing every step clearly. Correctly identifies and rejects the negative root with a clear real-world justification. Verifies the answer by substitution. Compares both methods thoughtfully.	Solves correctly using at least one method with mostly complete working. Rejects the negative root and states the correct dimensions. May not verify by substitution or may not fully compare the two methods.	Attempts a solution method but makes procedural errors (e.g., incorrect factors, arithmetic errors in the formula). May state both roots without rejecting the negative one, or may not connect the solution back to the hall's dimensions.
Graphical Understanding — Connecting the equation to the parabola	Accurately calculates and labels roots, y-intercept, and vertex. Sketch clearly shows an upward-opening parabola with all key features. Correctly identifies and explains the region where $f(w) < 0$ and interprets it meaningfully in the context of the hall's minimum width requirement.	Sketch includes roots and at least one other key feature. Identifies the region between the roots as where $f(w) < 0$ and gives a reasonable interpretation for the community context. Minor labelling or calculation errors may be present.	Sketch is attempted but key features are missing, mislabelled, or incorrectly calculated. Shows limited understanding of how the graph connects to the solutions of the equation or to the hall problem.
Real-World Application — Extending to a new Kenyan community context	Proposes a clearly described, realistic Kenyan community problem that genuinely requires a quadratic model. Correctly defines a variable, writes accurate expressions, and produces a valid quadratic equation in standard form. The context is specific and locally relevant (e.g., farming, construction, resource planning).	Proposes a relevant Kenyan context and attempts to model it with a quadratic equation. The variable is defined and the equation is mostly correct, though there may be minor errors in setting up expressions or reaching standard form.	Proposes a context but the mathematical modelling is weak — the equation produced may be linear, incorrectly formed, or not clearly connected to the described situation.
Communication and Mathematical Reasoning — Clarity, vocabulary, and reflection	Written explanations are clear, well-structured, and use correct mathematical vocabulary throughout (roots, discriminant, standard form, vertex, parabola, reject, verify). The reflection in Section 5(d) is thoughtful and makes a genuine, specific connection between the mathematics learned and its value in real Kenyan community and professional contexts.	Explanations are mostly clear and use some correct vocabulary. The reflection acknowledges the value of quadratic equations over trial-and-error, though the connection to wider Kenyan contexts may be general rather than specific.	Explanations are unclear, incomplete, or use informal language with limited mathematical vocabulary. The reflection is superficial or merely restates what was done without explaining why it matters.